Ontologies: The Keystone of AI

Let's talk about ontology for a second. So what is ontology? So ontology is, at its root, really a philosophical thing. It's a state of being.

But in AI, ontology really is a structured framework that's domain-specific. So a structured framework, what does that mean? So it's basically a way for human beings to communicate with computers and inform the computers about relationships, things that we know. And again, it's domain-specific.

So a stock in the financial market is different than a stock in retail, SKU, is different than a stock in agriculture, cattle, right? So we inform the different pieces of relationships. We inform the computer about that. And these things are nodes.

So nodes are objects. An example of that, let's say, there's the airline industry. And in the airline industry, you've got airports, you've got airlines, you've got different types of airplanes, passengers, et cetera.

And so what's the relationship? United Airlines flies from Francisco to Chicago. So that's a flight. We tell it the relationship, and we connect the different objects together.

And it forms a knowledge graph. It forms something that informs the computer about the relationships. A lot of companies that we talk to use LLMs and AI over top of relational databases.

Not so good. I mean, it's good. It's data analytics, right? And LLMs are great, and they can return some good results.

But they lack a lot of things. They lack a lot of insights that you can gain from an ontology. So think about it this way.

It's almost like a brain, right, where it's all connected and memories go through neurons. That's kind of what an ontology is. And the computer needs that to really understand how these different relationships manifest themselves.

So this relational database, like I said, can provide kind of simple data analytic answers, whereas an ontology can provide super insightful answers. And I think these are two really important stats that are out there in the market today. And let me tell you, they're probably not using ontologies in the failed projects.

99% of all AI projects have data issues. And those data issues oftentimes stem from lacking the relationship. So if you think about an Excel spreadsheet, inside the spreadsheet is the data.

But the rows and columns are the metadata. The tab is metadata. The title of the spreadsheet is the metadata.

You need all of that metadata to be absorbed into an ontology. And that allows the computer to then understand those relationships. And 99% of all AI projects have data issues, primarily because the data is not scrubbed and cleaned up front.

It's a different thing than ontologies, right? But the bottom line is data has to be used effectively to get effective answers out, right? So here's an example of the ontology for an airline. Right? And so what you can see is you've got luggage, passenger. Think about it as a tree.

I think that's a really good analogy, where you're in an airline. You're in a domain, the trunk. And then off are different branches to the ontology.

And so let's say there is a person, right? A passenger. Passenger has certain attributes. They live in a certain area.

That person has a family. They bring certain luggage. They have certain miles.

So that's one branch of the tree. Then another branch of the tree could be the airline. The airline owns 747s.

It owns MD-80s. It owns whatever. And so then you look at each of those planes.

And they have ages. So you can see where the ontology goes down, branches off. Ontology doesn't stop with the first implementation.

And ontologies, by the way, can be created pretty rapidly. We can create an ontology in a day, right? And then it's starting. That's the trunk.

But it's the refinement of the ontologies that really become important. And over time, using other ontologies, combined with your ontology, can create even a smarter machine. And there are a lot of industry-specific ontologies.

So like in the healthcare industry, there's something called SNOMED. That's a massive ontology maintained by groups of healthcare engineers, ML engineers. And it's a shared ontology.

And a lot of these industry-specific, domain-specific ontologies are shared across industries. But some of the value, the true value, is developing it yourself. If you're able to create an ontology for XYZ Bank, your bank is different than somebody else's bank.

At the root, or at the core, at the trunk, it's the same, but those branches are different. So it becomes your intellectual property. It becomes something that you can make as a competitive advantage.

You can start to refine things to give your AI more insights, smarter insights. So without this ontology, again, you can use a relational database, and you'll get some answers, and it's data analytics, but you won't get what I'm talking about, which are deep insights. So why do I say it's the keystone to AI? So keystone, you know, Roman times, it's basically an arc, right? And the center stone is the keystone.

All the stones lean on each other up to the keystone. If you pull the keystone out, the whole arch falls. So we believe, I believe, that if you don't have an ontology, especially in the future, as AI gets better, more advanced, so we're in gen AI, a lot of that's being used today, but agentic is right around the corner.

Agentic AI needs an ontology, right? And then if you can make that ontology better, smarter, et cetera, you're gonna have better results. So here's kind of the keystone, though, back to that, is if you have AI and you're using an ontology, you're able to do certain things and understand certain things. Natural language processing, a jaguar is an animal, not a car.

You can, an airline, you have one birthday and 50 flights, not 50 birthdays in one flight. So all of this stuff is available to you as you start to use these ontologies, but the number one thing that's very important in most industries that you don't get through using AI on a relational database, but you do with an ontology, is traceability. You get zero traceability through relational database and AI.

If you use an ontology, you get traceability all the way back to the decision, to the source data. Where is that important? Defense, finance, healthcare, manufacturing. We talk to a lot of companies.

I mean, I probably talk to three or four new prospective clients a week. Oh yeah, we're using AI. We're great, we've been doing it for two, three years.

Well, what kind of ontology do you use? Yeah, so we're good at AI. We're really good at it. We've been doing great stuff.

We get great results. Well, what are you using for your ontology? What tools did you use to develop it? What's an ontology, right? And so it's prevalent. I mean, I'm gonna say 90% of the companies I talk to, a $15 billion hospital network, a $20 billion manufacturer, they're not using ontologies.

And so it's definitely a chorus of ours, and there are a lot of other things that we could talk about, data governance and all that. But the ontology has to be used, and it has to be used at the beginning of the process. Data cleansing is gonna be important, and then you've got all the security concerns, DLP and all that, wrapped around it.

But the real key is that the ontology, if built properly at the beginning, enables your evolution, it gives you traceability, it provides a platform for growth, it provides a platform for agentic and future AI. There are a lot of tools out there, and I would highly recommend anybody who's considering building, extending AI to look at some of those tools and build a proper ontology. You'll get a competitive advantage.

You'll be able to actually create intellectual property, and you'll be able to build future state AI.